DTM-99: Best Practice Summary and Health-Check Checklist

> **Series:** DTM — Dynatrace Managed | **Reference:** 99 — Best Practice Summary and Health-Check Checklist | **Created:** September 2026 | **Last Updated:** 11/10/2026
>
> **Dynatrace Classic only.** This series applies to **Dynatrace Managed**, the self-hosted, on-premises deployment. None of it applies to Dynatrace SaaS.

Overview

This notebook puts the cluster-health notebooks on one page. DTM-10 introduced a five-layer health model for a self-hosted Dynatrace Managed cluster (node and process, storage, capacity, connectivity, lifecycle), and DTM-11 to DTM-16 covered one layer each. It also gathers the design and configuration checks from DTM-01 to DTM-05. This notebook turns them into something to run: a checklist per layer, one table of every signal that never reaches your mailbox, a review cadence, the numbers worth knowing, the anti-patterns that cause most of the trouble, and the gaps the documentation leaves.

Use it as the operating reference, and the earlier notebooks for the detail behind each line. For a one-off review of a cluster, with a rating for every check and a list of the evidence to request, use DTM-17.

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Prerequisites

Requirement	Details
Dynatrace deployment	A self-hosted **Dynatrace Managed** cluster. Nothing in this series applies to Dynatrace SaaS.
Access	Cluster Management Console (CMC) administrator access; root (or `sudo`) on the cluster nodes; a cluster API token with `ServiceProviderAPI` for the external checks
Query language	None. Managed has no Grail, so this series has **no DQL cells**
Read first	DTM-10. This notebook summarizes DTM-10 to DTM-16, and the design and configuration checks of DTM-01 to DTM-05, and points back to them
Version basis	Written against the Managed documentation read 10/01/2026

1. Short Answer

1. **Node and process.** Every node `RUNNING`, every service up. The cluster survives one node: *"Dynatrace Managed continues to operate after the loss of one node."* (From five nodes up, a Dynatrace blog allows two.)
2. **Storage.** Cassandra under 2 TB per node with every node `UN`; Elasticsearch `green`. Elasticsearch retention is a size lever; Cassandra's isn't.
3. **Capacity.** One-third headroom, and adaptive load reduction that never lasts 15 minutes or more.
4. **Connectivity.** The cluster kept internal behind Cluster ActiveGates, and a working Mission Control link.
5. **Lifecycle.** Backups that run, a restore you could actually perform, and a supported version.

Across all five, the most important habit is reading CMC **Events**. Many of the serious signals never email you (§4).

> **Sources:** Single-cluster high availability (DT docs), Configure Cluster event notifications (DT docs).

2. The Five-Layer Health-Check Checklist

Layer 1 — Node and process (DTM-11)

Check	How	Healthy
Every node present and running	`GET /api/v1.0/onpremise/cluster`, or CMC **Deployment status**	Every node's `operationState` is `RUNNING`; node count = cluster size
Every service up	`dynatrace.sh status` and `check` on each node	*"All services are OK"*, *"All rules are active."*, *"All processes are OK"*
Same version everywhere	The same API call	Identical `buildVersion` on every node
Clocks in sync	Host NTP monitoring	All nodes *"Synchronize with NTP"* and *"Be in the same time zone"*

Layer 2 — Storage (DTM-12, DTM-13)

Check	How	Healthy
Cassandra ring	`cassandra-nodetool.sh status`	*"Verify that all nodes display UN before proceeding to the next step."*; `Load` roughly even
Metrics store size	`du -sh` on each node	*"Keep the Long-term Metrics Store below 2 TB per node."*
Elasticsearch	`_cluster/health`	`"status" : "green"` (`yellow` only on a one-node cluster)

Check	How	Healthy
Retention settings	CMC → Environments → Storage settings	Each environment above the 35-day user-session default is a deliberate choice

Layer 3 — Capacity (DTM-15)

Check	How	Healthy
Headroom	Your load vs. the sizing table	*"Plan for a processing capacity one-third higher than typical utilization."*
Load reduction	CMC Events	Brief only. *"consistent use for intervals of 15 minutes or longer can impact the accuracy of your monitoring data"*
Ingest limits	CMC → Environments → Cluster overload prevention	Updated after the last hardware change

Layer 4 — Connectivity (DTM-14)

Check	How	Healthy
Mission Control link	CMC → *Check Mission Control connection*	Connected. Under classic licensing a lost link starts the clock: *"If a connection outage to Mission Control lasts longer than 14 days (seven days for free trial accounts), the Managed Cluster disallows the ability to exceed the license limit (overages)."* Under DPS, usage is sent once the connection returns
Load balancer	Its health checks	Every node checked at `/rest/health`; targets match the current nodes
Cluster ActiveGates	CMC → Deployment Status → ActiveGates	None *Unknown*; none flagged as behind by more than five versions
Certificates	Your certificate inventory	Expiry tracked outside the cluster

Layer 5 — Lifecycle (DTM-16)

Check	How	Healthy
Backups running	CMC **Home** → *Last backup*	Recent; no backup-problem events
Backup mount	`fstab` or your disk tool	Persistent. Otherwise *"If the shared file system mount point isn't available on system boot, Dynatrace won't start on that node."*
Restore readiness	Your restore kit	Exact installer version, node inventory, seed choice, matching UID:GID — and a step to re-enable backups, which a restore turns off
Version	CMC user menu vs. the release table	Inside "Currently supported"

> **Sources:**
>
> - Get cluster information about known cluster nodes (DT docs), Start/stop/restart a node (DT docs), Hardware requirements (DT docs)
> - Start/stop/restart a cluster (DT docs), Single-cluster high availability (DT docs), Adaptive traffic management for Managed (DT docs)
> - Mission Control proactive support (DT docs), Set up load balancer (DT docs), Update Cluster ActiveGate (DT docs)
> - Backup and restore a cluster (DT docs), Cluster Management Console (DT docs), Dynatrace Managed release notes (DT docs)

3. The Design and Configuration Review

The health checklist runs weekly. These checks run once when a cluster is built, and again after any change to its layout, its sign-in or its agents. Each row points to the notebook with the detail and the sources.

Area	Check	Good looks like	Notebook
Nodes	Node count, hardware, hosts	3+ nodes, identical hardware, one node per dedicated host, ≤ 10 ms apart, NTP and one time zone	DTM-01 §3
Availability	What the cluster survives	A chosen model — single cluster, rack awareness or Premium HA — written down with the failure it covers	DTM-01 §4
Headroom	Capacity with a node gone	Processing capacity one-third above typical load; both per-node limits (host units, Full-Stack OneAgents) checked	DTM-01 §2, DTM-15
Storage	Volume layout	Binaries and data on separate mount points, one volume per data type, LVM, equal sizes, antivirus exclusions, nothing on NFS, CIFS or EFS	DTM-01 §5
Operating system	Support runway	Every node on the same supported version, none on a version with an announced end date	DTM-01 §6
Properties	Customizations survive updates	Changes in `custom.settings` on every node, copy under change control	DTM-02 §2
Email	Alerts reach people	Own SMTP server with SSL/TLS or STARTTLS required, test message received, recipients and emergency contacts set	DTM-02 §3
Mission Control	Support relationship	Reporting preferences and remote-access scope and role chosen deliberately	DTM-02 §4
Sign-in	Local accounts	LDAP or SSO connected; 12-character minimum; concurrent-session limits and automatic sign-out set	DTM-02 §6, DTM-03 §4
Privileges	Root elevation	Agreed elevation method; Dynatrace service user's password expiry monitored	DTM-02 §7
Backups	Turned on	Off-site NFSv4 target, persistent mount, scope chosen	DTM-02 §9, DTM-16

Area	Check	Good looks like	Notebook
Environments	Boundaries and limits	Separate environments only where limits, retention, settings or visibility differ; quotas and retention set on purpose	DTM-03 §2–§3
Access	Least privilege	Small cluster-admin group; sensitive permissions narrow; restriction granted on management zones; classic permissions removed where policies replace them	DTM-03 §6–§7
Break-glass	Administrator account	Installation administrator stored as a break-glass credential, not used day to day	DTM-03 §5
ActiveGates	Roles and redundancy	One purpose each; Environment ActiveGates in every OneAgent segment; 2+ per data center per job; `dnsEntryPoint` set behind load balancers	DTM-04
ActiveGate updates	Currency	Update mode and target version set for Environment ActiveGates; Cluster ActiveGates within five versions	DTM-04 §7
OneAgents	Install-time choices	Host groups and network zones designed before rollout and set on the command line; fallback mode chosen per zone	DTM-05 §3–§4
OneAgent updates	Policy	Host groups as update rings in shared update windows, restarts planned, statuses watched if automatic updates are off	DTM-05 §5
CPUs	x86-64-v2	Hosts and VM templates below the baseline identified	DTM-05 §6

4. Signals That Never Email You

Cluster events appear in the CMC **Events** list, and the docs' event table gives each one an email column. These are the ones in this series whose email column is **No**, grouped by layer. None of them will reach you unless someone reads the Events list:

Layer	Event (as the docs print it)	Severity
1	*" is down."*	SEVERE
1	*"Heap memory: Server %d started memory emergency mode."*	SEVERE
1	*"Insufficient system privileges on %s."*	SEVERE
1	*"Node cannot read and write to directory: '%s'."*	WARNING
1	*"Server time of server %d is out of sync. Time difference %d milliseconds. Please enable NTP on all cluster nodes."*	not shown
2	*"Long-term Metrics Store size exceeds recommended 2 TB on %s."*	WARNING
2	*"Cassandra node connection lost (%d times in the last hour)."*	WARNING
2	*"Elasticsearch storage service on your Dynatrace Managed cluster might be overloaded!"*	INFO
2	*"Elasticsearch log queue is full."* / *"Elasticsearch log storing failed."*	WARNING
3	*"Cluster traffic control: OneAgent monitoring was disabled on recently connected hosts to avoid cluster overload."*	SEVERE
4	*"A cluster node can't receive OneAgent traffic."*	SEVERE
4	*"SSL certificate expired."* / *"Your SSL certificate will expire soon."*	SEVERE / WARNING
4	*"ActiveGate (host=…) lost connection to cluster."*	INFO
5	*"Cassandra backup problem."* / *"ElasticSearch backup problem."*	SEVERE / WARNING or SEVERE

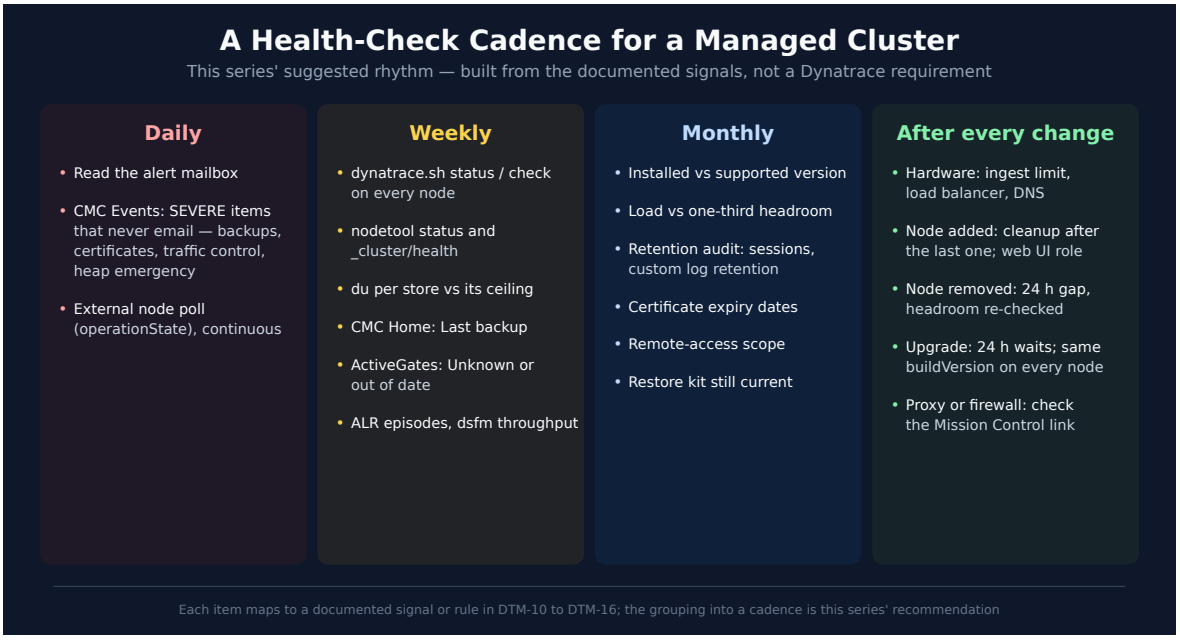
Layer	Event (as the docs print it)	Severity
5	*"Dynatrace update %s download failed."* / *"Post-Update data migration failed"* … / the cluster *update failed* event	SEVERE / SEVERE / WARNING or SEVERE

For contrast, these **do** email: *"Node is down - %s."* (*"if happened outside of the upgrade procedure"*), *"Host is down."*, and *"There is lack of connection to Dynatrace Mission Control."*. So does *"Your Dynatrace Managed cluster should be scaled-out!"*, and *"Backup has been disabled because the configuration is no longer supported."* The *Configurable* ones are insufficient disk space, an undersized node (*"Your Dynatrace Managed cluster node {0} is undersized!"*), the 4 TB metrics ceiling, retention truncation and adaptive load reduction. Whether they email by default is contradicted on the events page itself: its default example sets truncation and adaptive load reduction to `sendEmail: false`. Set all of them explicitly (DTM-10 §6.2).

> **Sources:** Configure Cluster event notifications (DT docs), Event notification reference (DT docs) — Email notification column, read 11/10/2026.

5. A Review Cadence

The documentation sets no review schedule. This is the series' suggestion for turning the checklist into a routine. Each item comes from a documented signal or rule, but the grouping is ours.



The "after every change" column is the one teams most often skip. Each item in it is something a hardware, node, upgrade or network change leaves behind for someone to do.

6. Numbers Worth Knowing

Number	What it is	From the docs
3	Production minimum	*"Production environments require a minimum three-node Managed Cluster for reliability and data redundancy."*
42	Maximum nodes	*"Dynatrace Managed supports up to 42 Cluster nodes."*
3 / 2	Copies: metrics, user sessions / log events	*"Replication factor is set to three."* (user sessions); *"Replication factor is set to two."* (logs)
2 TB / 4 TB	Metrics store per node: recommended ceiling / severe cluster event	*"Keep the Long-term Metrics Store data below 2 TB per node."*; *"a severe cluster event when it exceeds 4 TB."*
⅓	Headroom	*"Plan for a processing capacity one-third higher than typical utilization."*
10 ms / 100 ms	Latency between nodes / between Premium HA data centers	*"Have a network latency between nodes of 10 ms or less"*; *"Keep round-trip network latency at 100 ms or less."*
15 min	Sustained load reduction that costs accuracy; Premium HA failover trigger	*"consistent use for intervals of 15 minutes or longer"*; nodes *"down for 15 minutes"*
24 h	Between node removals; after a download; between updates	*"allow 24 hours before removing any subsequent nodes"*; *"a 24-hour waiting period is required before an automatic update can run"*
14 days	Mission Control outage before overages stop (classic)	*"lasts longer than 14 days (seven days for free trial accounts)"*

Number	What it is	From the docs
72 h	Premium HA automatic repair window	*"PHA automatically repairs network disconnections of up to 72 hours between DCs."*
5 GB	Free space needed to update	*"You have at least 5 GB of available disk space on the partition where Dynatrace Managed is installed."*
4 weeks	Release cadence	*"typically released every four weeks"*

> **Sources:**
>
> - Hardware requirements (DT docs), Data retention periods (DT docs), Single-cluster high availability (DT docs)
> - Multi-data center high availability (DT docs), Multi-data center failover (DT docs), Adaptive traffic management for Managed (DT docs)
> - Remove a cluster node (DT docs), Update a cluster (DT docs), Mission Control proactive support (DT docs)

7. Anti-Patterns

Anti-pattern	What the docs say happens
Trusting the mailbox alone	Backup failures, certificate expiry, traffic control and heap emergencies never email (§4)
Nodes of different sizes	Every node must *"Have the same hardware configuration"*; *"When adding CPUs or RAM, keep all nodes equally sized."*
Network file systems for live data	*"High-latency remote volumes such as NFS or CIFS are not supported for primary storage; NFS is sufficient for backups only."*
A backup mount that isn't persistent	*"If the shared file system mount point isn't available on system boot, Dynatrace won't start on that node."*
Restarting a 3+ node cluster with `dynatrace.sh`	*"For Managed Clusters with three or more nodes, go to Start/stop/restart a Managed Cluster."*

Anti-pattern	What the docs say happens
More than one node out at once	*"The loss of two or more nodes might affect Managed Cluster performance and availability"*
Running at capacity	With no headroom, a node failure moves its traffic onto nodes that can't absorb it: *"If a node fails, the NGINX load balancer automatically redirects all OneAgent traffic to the remaining working nodes."*
Exposing the cluster directly	*"Exposing the Managed Cluster directly to external networks is not recommended for security reasons."*
Forgetting the change follow-ups	*"Update the Managed Cluster ingest limit via Cluster Management Console or REST API whenever hardware changes."*; for a load balancer, *"This list needs to be updated after node join or removal."*
Assuming a backup is a full copy	*"Transaction storage data is not backed up, so after a restore you might see gaps in deep monitoring data"*

> **Sources:**
>
> - Configure Cluster event notifications (DT docs), Hardware requirements (DT docs)
> - Backup and restore a cluster (DT docs), Start/stop/restart a cluster (DT docs)
> - Single-cluster high availability (DT docs), Managed deployments (DT docs), Set up load balancer (DT docs)

8. What the Documentation Does Not Say

Each notebook ends with its own gaps. Taken together, they say where to rely on your own monitoring rather than a documented threshold.

Not documented, as read by 10/01/2026:

Area	Gap	Notebook
Node state	An official list of `operationState` values (the docs show only `RUNNING`; a community-posted list of 12 exists)	DTM-10, DTM-11

Area	Gap	Notebook
Node operations	A procedure for restarting one node of a 3+ node cluster	DTM-11
Memory	What "memory emergency mode" does to processing	DTM-11
Storage	Dynatrace-set disk thresholds (community KBs cite Elasticsearch's upstream watermarks and Cassandra's compaction headroom)	DTM-12, DTM-13
Storage	Where Davis problems and events are stored (a 2019 community answer says Elasticsearch)	DTM-10, DTM-13
Self-monitoring	`dsfm:` metrics for the stores' own health (the `dsfm:active_gate.*` keys are now covered in DTM-14)	DTM-12, DTM-13
Capacity	A number behind "sufficient capacity" or behind adaptive load reduction	DTM-15
Connectivity	What stops working when the cluster certificate expires (community answers: monitoring continues, the UI warns)	DTM-14
Lifecycle	A way to rehearse a restore without disturbing production	DTM-16

Conflicts between pages:

Topic	The two statements	Series follows
Elasticsearch snapshot interval	Every 2 hours vs. every 2 days	2 hours, from the backup page (DTM-13)
Cluster ActiveGate inbound port	9999 vs. HTTPS on 443	Neither stated (DTM-14)
Mission Control health-check interval	Every 2 minutes vs. every 5 minutes	2 minutes, from the data-exchange page (DTM-14)
Local self-monitoring	Every cluster vs. DDU licensing only	Both recorded; a Dynatrace blog sides with "every cluster" (DTM-10, DTM-15)
Premium HA copy count	Two copies vs. three copies in each DC	Unresolved; confirm with Dynatrace support (DTM-10)

> **Observed 10/01/2026:** this table consolidates the documentation-gap sections of DTM-10 to DTM-16;

> each row was searched for, and each conflict quoted, in the notebook named.

9. Beyond the Docs

The Managed documentation is the primary source for this series. These are the most useful additional resources, checked 10/02/2026. Community items are answers and articles on the Dynatrace community, some by Dynatrace staff. They are useful practice, not documentation, and several are years old.

Resource	Type	Useful for
ActiveGate self-monitoring metrics	Dynatrace docs	The 70 `dsfm:active_gate.*` keys (DTM-14)
Proactive self-monitoring for Dynatrace Managed	Dynatrace blog	Reading the capacity indicator (DTM-15)
Premium High Availability and turnkey disaster recovery	Dynatrace blog	Node-loss tolerance by cluster size (DTM-10)
Dynatrace Self-Monitoring (Managed)	Dynatrace Hub	A packaged capacity view (DTM-15)
dtmgd and dynatrace-managed-mcp	Dynatrace open source (Apache-2.0)	A CLI and an MCP server for Managed environments and the cluster API. The MCP server's documented token scopes are read scopes. `dtmgd` v0.5.0 and later also writes Settings 2.0 objects where a context's write level allows it; *"every context is `readonly` until you raise it"* (dtmgd README)
state of nodes	Dynatrace community	All 12 `operationState` values (DTM-11)
How to restart Dynatrace nodes on the cluster	Dynatrace community	The CMC per-node Restart (DTM-11, a recorded conflict)

Resource	Type	Useful for
Elasticsearch store usage differences	Dynatrace community KB	Elasticsearch watermarks (DTM-13)
Reserve disk space consumed	Dynatrace community KB	Cassandra compaction headroom (DTM-12)
Elasticsearch won't start	Dynatrace community KB	A missing backup mount stopping Elasticsearch (DTM-13, DTM-16)
Elasticsearch log queue is full	Dynatrace community KB	Log-ingest pressure (DTM-13)
Is Cassandra corrupted?	Dynatrace community KB	Spotting corrupt Cassandra tables (DTM-12)

The community pages block scripted fetches; open them in a browser.

10. Summary and Where to Go Next

A healthy Managed cluster is five things at once. Its nodes and services run. Its stores stay under their ceilings with every replica in place. It has a third more capacity than it uses. Data can reach it and it can reach Mission Control. And it can be restored and stays supported. Most of what goes wrong announces itself only in CMC Events, so the routine matters more than any single alert: daily for what doesn't email, weekly for the node, store and backup checks, monthly for versions, headroom and retention, and after every change for the follow-ups.

The cluster-health notebooks:

Notebook	Layer
DTM-00: The Dynatrace Managed Journey	Series map
DTM-10: Managed Cluster Architecture and Health Model	The model
DTM-11: Server Nodes and Processes	1 — Node and process

Notebook	Layer
DTM-12: Cassandra Metrics Store Health	2 — Storage
DTM-13: Elasticsearch Store Health	2 — Storage
DTM-14: Cluster ActiveGates and Mission Control Connectivity	4 — Connectivity
DTM-15: Capacity and Scaling	3 — Capacity
DTM-16: Backup, Upgrade, and Disaster Recovery	5 — Lifecycle
DTM-17: Running a Cluster Health Assessment	All layers, as a one-off rated review

The design-and-configure notebooks:

Notebook	Covers
DTM-01: Cluster Design Best Practices	Node count, availability model, storage, hosts
DTM-02: Cluster Configuration Best Practices	Properties, email, Mission Control, sign-in, privileges, backups
DTM-03: Environment Best Practices	Environments, identity sources, permissions
DTM-04: ActiveGate Deployment on Managed	ActiveGate roles, placement, zones, updates
DTM-05: OneAgent Deployment on Managed	Routing, install-time choices, update policy

Moving off Managed? The M2S series (Managed to SaaS Migration) is the next step. A healthy source cluster is the best starting point for a migration.

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.